Exercise Sheet 01

Priya's Inbox

House rules.

- One file per exercise, in `my_work/`: `ex1_pay_maths.py`, `ex2_names.py`, …
- Start each file by typing the **Given** lines. Then write the rest yourself.
- Run it (`python my_work/ex1_pay_maths.py`) and compare with the **expected output**, character by character.
- Stuck for more than 10 minutes? Look at the cheat sheet, then skip and come back later.
- Hand in: commit and push to `lesson-01`, then open a pull request. Tell Lars which ticket was the most annoying.

Exercise 1 Pay maths

10 min · `ex1_pay_maths.py`

> **From Priya (Payroll):** *"MACRO-9000 crashed again over the weekend. Run-time error '13', obviously. Can you redo the basic pay maths in Python? Cheers, P."*

GIVEN

```
annual_salary = 32500
weekly_hours = 37.5
part_time_hours = 22.5
holiday_days = 17
raise_percent = 3
years = 3
```

Calculate and print, each rounded to 2 decimal places where it makes sense:

(a) the monthly salary;

(b) the hourly rate (52 weeks a year);

(c) the salary of someone doing the same job for `part_time_hours` a week;

(d) how many full weeks and leftover days `holiday_days` are (use `//` and `%`);

(e) the salary after `years` yearly raises of `raise_percent` %, each on top of the last.

EXPECTED OUTPUT

```
Monthly salary: 2708.33
Hourly rate: 16.67
Part-time salary: 19500.0
Holiday: 2 weeks and 3 days
Salary after raises: 35513.63
```

Hint: "To the power of" is `**` in Python. If you use `^`, you'll meet it again in the bug hunt.

Exercise 2 Names and text

10 min · `ex2_names.py`

> **From Mo (IT):** *"The staff directory export is a mess. Dave's macro 'cleaned' it by deleting every second row. Please be gentler than Dave."*

GIVEN

```
raw_first = "  jANE "
raw_last = "smITH  "
raw_id = "  emp-0042 "
```

(a) Clean the names (no spaces around them, proper capitals) and print the full name.

(b) Print the initials with dots.

(c) Print the company email: *first.last*@brightwell.co.uk, all lowercase.

(d) Clean the ID: no spaces, uppercase, no dash.

(e) Siobhan's colleague is called `"Anne Marie"` `"Brown"`. Print her email, without the space.

EXPECTED OUTPUT

```
Jane Smith
J.S.
jane.smith@brightwell.co.uk
EMP0042
annemarie.brown@brightwell.co.uk
```

Exercise 3 A payslip Gerald can love

10 min · `ex3_payslip.py`

> **From Gerald (MD), Friday 16:55:** *"The payslips look 'a bit 1997'. Commas! Pound signs! Things lined up! Have a lovely weekend. G."*

GIVEN

```
name = "Jane Smith"
month = "March"
annual_salary = 32500
pension_rate = 0.05
tax_rate = 0.20
```

Basic pay is a twelfth of the salary. Pension is 5 % of basic pay. Tax is 20 % of (basic pay minus pension). Net pay is what's left. Print exactly this:

EXPECTED OUTPUT

```
PAYSLIP: Jane Smith, March
---------------------------
Basic pay         £2,708.33
Pension             £135.42
Income tax          £514.58
---------------------------
Net pay           £2,058.33

Pension rate: 5.0%
Annual salary: £32,500
```

Hint: Each line is 27 characters: the label left-aligned in 15, the amount right-aligned in 12. Make the money text first: `basic_text = f"£{basic:,.2f}"`, then `f"{'Basic pay':<15}{basic_text:>12}"`. And `"-" * 27` draws the line.

Exercise 4 Text that should be numbers

10 min · `ex4_conversions.py`

> **From Priya:** *"The HR system exports EVERYTHING as text. Apparently '£32,500.00' is not a number. MACRO-9000 says 'Type mismatch'. I say 'please'."*

GIVEN

```
salary_text = "£32,500.00"
hours_text = " 37.5 "
bonus_text = "12.5%"
member_text = " Yes "
employee_number = 42
```

(a) Turn the first three into numbers and `member_text` into `True` / `False`. Print each value together with its `type()`.

(b) Build the employee ID: `EMP` followed by the number padded to 4 digits.

(c) Print the bonus in pounds (salary × bonus rate).

EXPECTED OUTPUT

```
32500.0 <class 'float'>
37.5 <class 'float'>
0.125 <class 'float'>
True <class 'bool'>
EMP0042
Bonus: £4,062.50
```

Exercise 5 Yes or no

5 min · `ex5_true_false.py`

> **From Siobhan (Reward):** *"Just yes or no, please. Dave's version once answered 'maybe'."*

GIVEN

```
weekly_hours = 30
months_of_service = 14
rating = 4
salary = 33500
band_min = 25000
band_max = 32000
```

(a) Print four answers. **Full time**: 35 hours or more. **Bonus eligible**: at least 12 months *and* rating 3 or more. **Within band**: salary between min and max, limits included. **Needs review**: not within band.

(b) Before running it again: if `weekly_hours = 37.5` and `salary = 30000`, what will it print? Write your guess as a comment at the bottom of the file, then try it.

EXPECTED OUTPUT

```
Full time: False
Bonus eligible: True
Within band: False
Needs review: True
```

Exercise 6 LEFT, RIGHT, MID

10 min · `ex6_slicing.py`

> **From Mo:** *"In Excel I'd use LEFT/MID/RIGHT. Dave used 400 lines of VBA. Please use fewer."*

GIVEN

```
company = "Brightwell"
ref = "SALES-1203"
```

(a) Print the Python versions of `LEFT(company, 6)`, `RIGHT(company, 4)` and `MID(company, 3, 4)`.

(b) Split `ref` at the dash: print the department, then the number as a real number.

(c) Print the number plus one, to prove it really is a number now.

EXPECTED OUTPUT

```
Bright
well
ight
SALES
1203
1204
```

Hint: Excel counts from 1, Python from 0. `ref.find("-")` tells you where the dash is. Your code should still work if the ref were `"HR-0007"`. Try it!

Exercise 7 The pay gap in one sentence STRETCH

15 min · `ex7_pay_gap.py`

> **From Gerald:** *"The board wants our pay gap 'in one sentence a human can read'. Meeting's Thursday. No pressure. (Some pressure.) P.S. Biscuit's walk log is in decimal hours and nobody understands it."*

GIVEN

```
male_hourly = 20.00
female_hourly = 17.40
walk_hours = 1.75
```

The UK definition: **gap % = (male − female) / male × 100**, rounded to 1 decimal. Pence per pound = female / male × 100, rounded to a whole number.

EXPECTED OUTPUT

```
Women earn 87p for every £1 that men earn (gap: 13.0%).
Biscuit walked for 1h 45m.
```

Hint: `int(1.75)` is `1`. The minutes come from what's left over. `{minutes:02d}` pads to two digits.

Bug hunt Dave's Leftovers #1

15 min · `bughunt/payroll_bug.py`

Dave translated part of MACRO-9000 to Python and left. Since then the raise calculation crashes, and for three months nobody at Brightwell paid any income tax (HMRC has questions). There are **three** bugs: one crashes, one gives a wrong number silently, and one you'll only spot by comparing letter by letter.

EXPECTED OUTPUT ONCE FIXED

```
Salary after 2 years: £31,827.00
Omar Khan: gross £2345.50, tax £469.10, net £1876.40
```

Commit each fix separately, with a message explaining what was wrong, like a detective's report.